
Using Deep Learning to Detect Apple Leaf Disease

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8.1 INTRODUCTION

Kashmir, land rich in its culture and languages, is also rich in its vegetation. Its soil supports the growth of a variety of fruits, the apple being the most famous. Apple production brings in a lot of wealth. The apple tree originated in central Asia and is now cultivated worldwide. Its botanical name is *malus domestica* [1]. In Kashmir, a wide variety of apples are cultivated viz Delicious, Golden, Ambri, Chamboora, etc. Apples are grown in almost every district of Kashmir and generate annual revenue of Rs. 1500 crores [2]. The apple tree is, however, vulnerable to certain kinds of disease, which affect its leaves. The typical apple leaf diseases found in Kashmir are Alternaria leaf blotch, apple scab, and apple rot, as shown in Figure 8.1. These diseases render the whole orchard devastated, which is a big challenge for the farmers. These diseases damage the quality and decrease the quantity of crops. The traditional ways to diagnose the disease cause greater damage, as the methods are slow, labor-intensive, and nonreliable. These methods need experts with extensive knowledge and experience and are often inaccurate. The pesticide that is then used is also improper and becomes an economic and environmental burden.

Now it has become the need of the hour to change the traditional time-consuming methods to novel technology-based methods. The advanced technology has put machine learning in use, which automatically detects and classifies the disease of apple leaves. However, this method has a

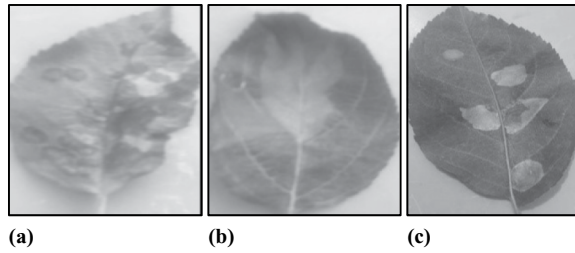


FIGURE 8.1

Typical apple diseases: (a) *Alternaria* leaf blotch, (b) apple scab, and (c) apple rot.

drawback, as it cannot be applied to raw data and requires preprocessing. Deep learning has now emerged as a better alternative, as it can be applied to raw data, and no extra step of preprocessing is needed. Further advancements in deep learning have led to the idea of a convolution neural network (CNN). It is particularly used for image classification. CNN comprises two components – viz, convolution and pooling used for feature extraction, and the classifier component, which predicts the output. In this study, various versions of CNN algorithms have been used for predicting and classifying different types of disease in apple leaves.

This chapter contains four sections. The first section is about related work done in related fields. The second section talks about experimental details. This section includes methods, model building, data set, and experimental environment. The third section is about analyzing the results using different performance evaluation metrics. The last section sums up the conclusion of the study.

8.2 RELATED WORK

Deep learning and machine learning techniques have been practiced on different disease predictions on different types of agricultural products. The research related to the identification and classification of plant leaf diseases is presented in this section. The remainder of this section gives a detailed explanation of the research undertaken in this field. The summarization of this section is given in Table 8.1.

Zhang et al. [3] introduced a new method of apple leaf disease identification. The model was based on the support vector machine (SVM) classifier. The dataset for the identification of diseased apple leaves contains 90

TABLE 8.1

Related Work Done in Related Fields

Author	Year	Description	Algorithm	Classification Accuracy (%)
Zhang et al. [3]	2017	Identification of apple leaf diseases	SVM	90
Dubey et al. [4]	2012	Identification and categorization of apple fruit disease	SVM	93
Arivazhagan et al. [5]	2013	Detection and classification of leaf disease	SVM	94
Brahimi et al. [6]	2017	Recognition of tomato leaf diseases	CNN	99
Khan et al. [7]	2021	Classification and identification of apple diseases	CNN	97.18
Jan et al. [8]	2020	Apple disease diagnosis for predicting apple scab and leaf/spot blight disease	MLP	99.1

images of healthy apple leaves, mosaic leaf, powdery mildew leaves, and rust leaves. This approach acquired an identification accuracy of more than 90%. S. R. Dubey et al. [4] developed a system for the detection and classification of apple fruit disease. In this system, image segmentation was done using K means clustering. The segmented images were then used to obtain the art features. Finally, the images are classified into apple blotch, apple rot, and apple scab using multiclass SVM. This model achieved an accuracy of 93%. Arivazhagan et al. [5] developed a software-based automatic recognition and classification of plant leaf disease. For the processing part, color conversion and segmentation were applied to images. The system used an SVM classifier for recognition and classification. An accuracy of 94% was achieved. Brahimi et al. [6] also applied a deep-learning (CNN algorithm) model for a similar study classifying tomato disease. The model used leaf images and achieved a classification accuracy of 99%. Khan et al. [7] used the concept of CNN to perform feature extraction from leaf images. The CNN model worked to predict the diseased leaves and acquired an accuracy of about 97%. Jan et al. [8] developed an outstanding apple disease recognition system to recognize leaf/spot blight and apple scab disease. The proposed system used

shape-based and low-level features of leaves to train a multilayer perception (MLP) model. The model achieves an accuracy of 99.1%.

8.3 PROPOSED METHODOLOGY

The methodology employed in this study is a structured approach designed to comprehensively address the challenges of predicting and classifying apple leaf diseases. These steps encompass acquiring an image dataset, performing image preprocessing, constructing a predictive model, evaluating the model's performance, and analyzing the results. The methodology is shown in the following flow chart (Figure 8.2).

8.3.1 Methods

The study has been conducted using a custom CNN, VGG 16, ResNet-50 and Inception V3. The pretrained networks VGG 16, ResNet-50, and Inception V3 are all powered by convolution neural techniques. The working of CNN, along with its mathematical modeling, is explained next.

CNN is a type of deep-learning algorithm that is mostly used for image classification [9]. The architecture of CNN contains many layers, and each layer performs a different function. The layers of CNN are connected to each other, and the output of a layer is input for the next layer [10]. Mainly, there are three layers in CNN – namely, the convolution layer, pooling layer, and fully connected layer.

8.3.1.1 Convolution Layer

The convolution layer is a key layer in CNN, whose function is feature extraction [11]. Feature extraction is done by using a filter or kernel. In the convolution layer, an activation/feature map is formed by the convolution operation between the input image and the kernel.

Equation 8.1 represents the convolution operation mathematically.

$$O_j = f \left(\sum_{i=1}^n a_i * k_{i,j} + b_j \right) \quad (8.1)$$

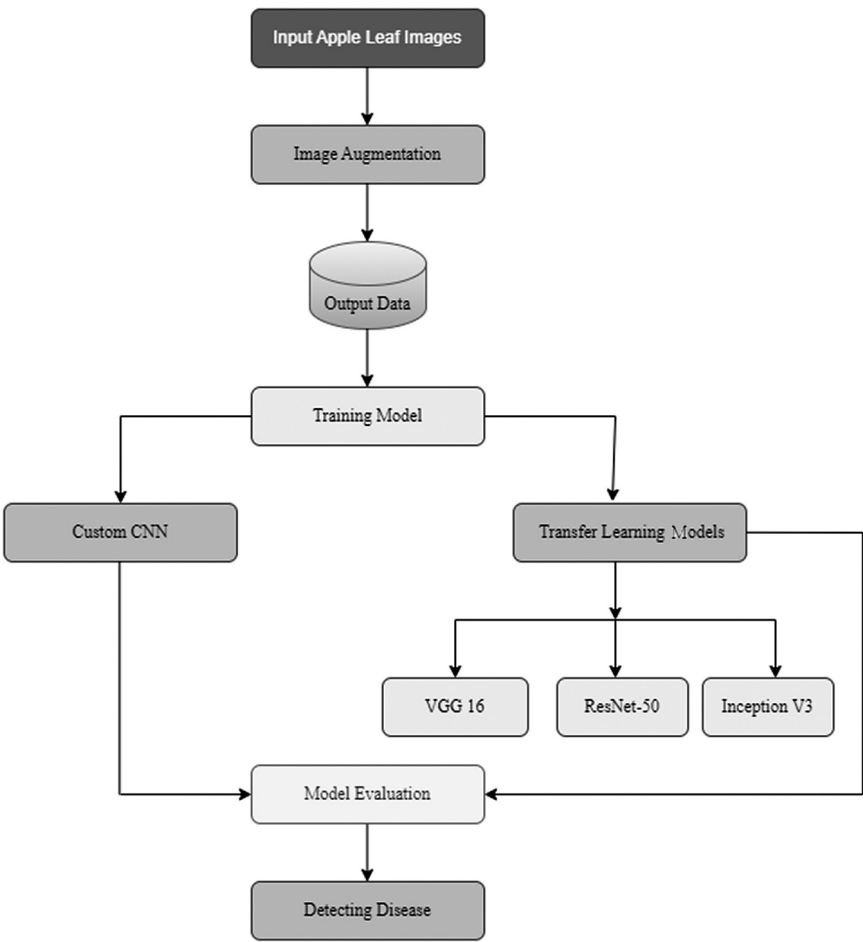


FIGURE 8.2
Methodological steps.

First of all, the input image a_i is convoluted with the kernel $k_{i,j}$; then the summation of convoluted operation is calculated, and a bias b_j is added with the resultant. Lastly, an activation function f is applied to the resultant, and the feature map O_j is produced.

Bias is a constant value that is used within an activation function in order to shift the position of the curve left or right to delay or accelerate the firing of a neuron [12].

The function of the activation function is to decide whether the neuron is fired or not. The most widely used activation is the Rectified Linear Unit (ReLU). The advantage of the ReLU function is that it does not activate all

the neurons simultaneously. The ReLU function does not fire the neuron if the input values are negative [13].

Equation 8.12 represents the ReLU function mathematically.

$$g(x) = \max(0, x) \quad (8.2)$$

8.3.1.2 Pooling Layer

The pooling layer is present after the convolution layer in CNN. The function of the pooling layer is to reduce the dimensions of the feature map. In pooling, the neighboring elements of convolution output are combined [11]. Average pooling and max pooling are the main pooling operations.

Average pooling is a pooling operation that filters the feature map and selects the average value from the filtered patch [14], mathematically represented in Equation 8.3.

$$f_{\text{avg}}(x) = \frac{1}{n} \sum_{i=1}^n x_i \quad (8.3)$$

Max pooling is a pooling operation that filters the feature map and selects the maximum value from the filtered patch [14]. Equation 8.4 gives the mathematical representation of this.

$$f_{\text{max}}(x) = \max(x_i) \quad (8.4)$$

8.3.1.3 Fully Connected Layer

After a number of convolution and pooling layers, there are a number of fully connected layers. The output (feature map) from the convolution and pooling layers are joined with each other to form a larger feature map. In another way, a fully connected layer flattens the input image so that it converts the multidimensional input image into a one-dimensional array (vector) [10]. Each fully connected layer has an activation function. Mostly for multiclass image classification, the Softmax activation function is used for the last fully connected layer. Softmax function converts the one-dimensional array into the probability distribution that sums up to 1 [15].

Equation 8.5 represents the Softmax activation function mathematically.

$$S = \frac{\exp^{z_i}}{\sum_{j=1}^k \exp^{z_j}} \quad (8.5)$$

8.3.2 Model Building

The CNN algorithms used in this study for model building are given in the next section:

8.3.2.1 Custom CNN

The proposed custom CNN is an 11-layer CNN containing 4 convolution layers, 4 max-pool layers, and 3 dense layers. The input size of 150×150 is passed to the 11 layers of the network. This model uses the ReLU activation function for hidden layers and the Softmax activation function for the output layer. Figure 8.3 shows the architecture of the model.

8.3.2.2 VGG16

VGG (visual geometric group), proposed by Zisserman et al. in the year 2014, is a 16-layer CNN working on 13 convolution layers, 5 max-pooling layers, and 3 dense layers that add up to 21 layers, out of which only 16 are weighted [16]. The architecture is shown in Figure 8.4. The input size of 224×224 is passed to 16 layers of the network [17].

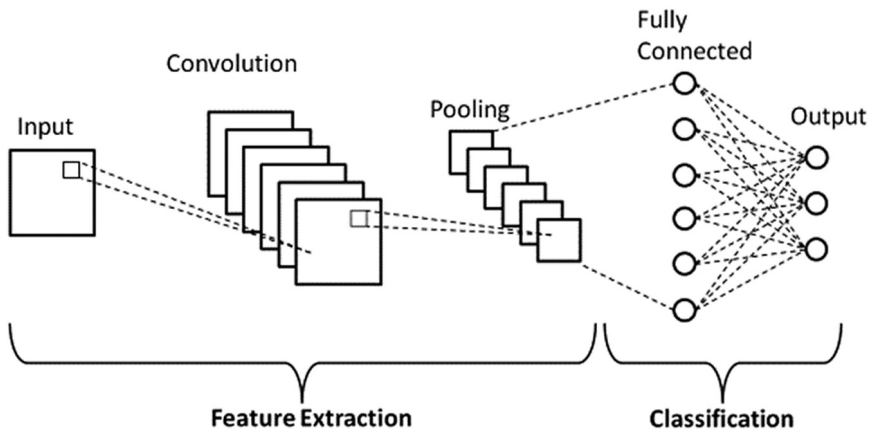


FIGURE 8.3

Architecture of custom CNN.

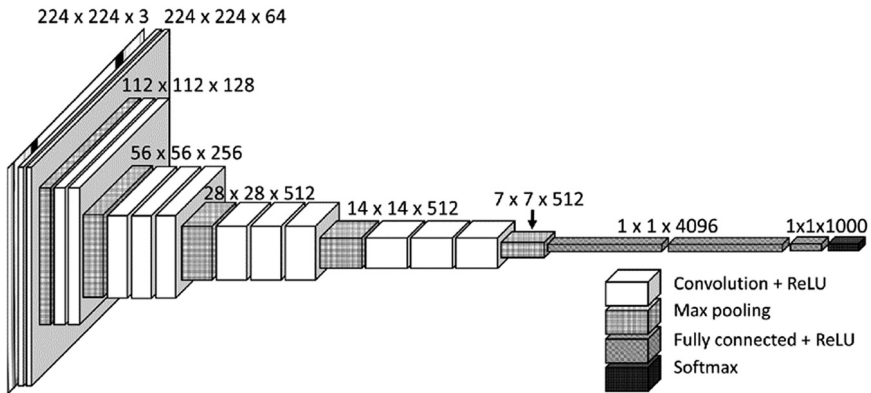


FIGURE 8.4
Architecture of VGG16.

8.3.2.3 ResNet-50

ResNet (residual network), proposed by Kaiming et al. in the year 2015, is a 50-layer CNN with 48 convolution layers, 1 max-pool layer, and 1 average pool layer [18]. The architecture of Resnet-50 is shown in Figure 8.5. The input size of 180×180 is passed to the 50 layers of the network [19].

8.3.2.4 InceptionV3

Inception v3 is proposed by Szegedy et al. in the year 2015 [20]. The inception V3 contains 5 convolution layers, 2 max-pooling layers, 11 inception modules, 1 average pooling layer, and 1 dense layer [21]. The architecture of Inception V3 is shown in Figure 8.6. The input size of 229×229 is passed to the 48 layers of the network.

8.3.3 Dataset

The dataset for this study was collected from the dataset repository Kaggle under the title “Kashmir Apple Leaf Disease” [22]. The data set contains 419 images of infected and healthy leaves. Table 8.2 contains the number of images for individual classes.

8.3.4 Data Augmentation

Data augmentation is a technique by which artificial data is generated from the dataset in order to decrease overfitting during the training of a model. In augmentation, geometric and color space transformations

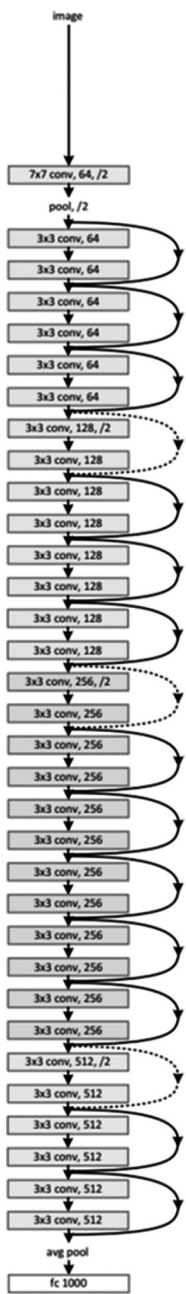


FIGURE 8.5
Architecture ResNet-50.

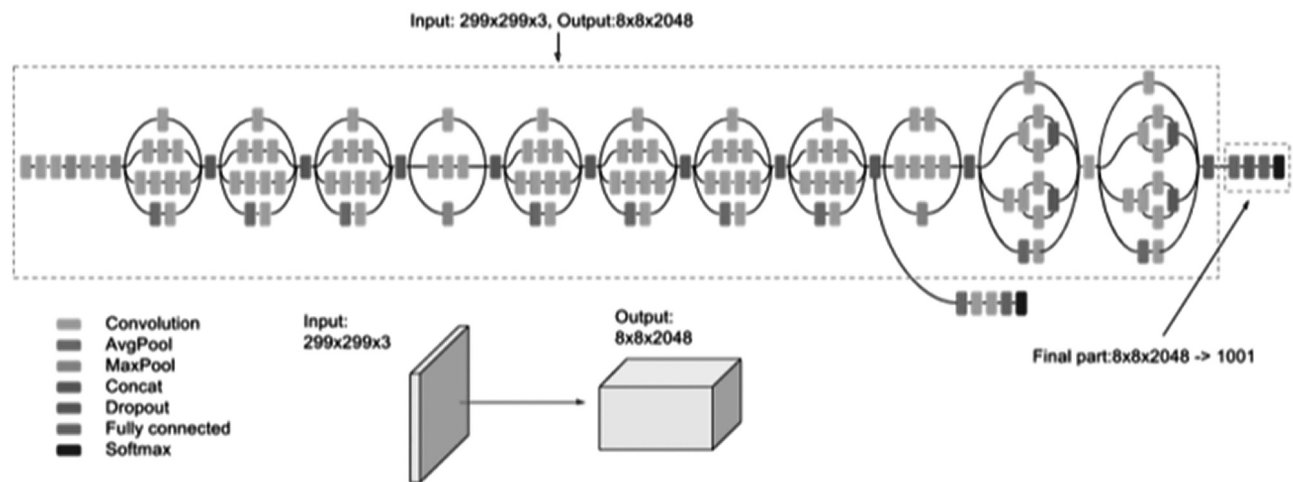


FIGURE 8.6
Architecture of inception V3.

TABLE 8.2

Number of Records in Dataset

Class	Total Number of Images
Healthy	46
Apple Rot	103
Alternaria leaf Blotch	111
Apple Scab	159

(flipping, resizing, cropping, brightness, and contrast) are used [23]. In this study, 419 image files of the dataset have been augmented to generate 2,000 files.

8.3.5 Experimental Environment

Coding for experimentation is performed within a Python environment called Google Colab – a product developed by Google Research. It enables the writing and execution of code directly through a web browser [24].

8.4 MODEL EVALUATION

Model evaluation refers to the methodology of evaluating the model’s performance. Several evaluation parameters are used for that purpose that are all based on the following outcomes of classification predictions: True Negative (TN), True Positive (TP), False Positive (FP), and False Negative (FN). Conventional performance evaluation parameters are accuracy, precision, recall, and F1-score [25].

- Accuracy: The proportion of predictions classified correctly by the model.

$$\text{Acc} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{TP} + \text{FP} + \text{FN}} \quad (8.6)$$

- Precision: The ratio of positive values classified correctly to all the positives.

TABLE 8.3

Evaluation Parameters

Algorithm	Classification Accuracy (%)	Precision	Recall	F1-Score
Custom CNN	80	0.78	0.77	0.77
VGG16	85	0.82	0.81	0.81
ResNet-50	81	0.79	0.77	0.78
InceptionV3	0.83	0.80	0.78	0.79

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (8.7)$$

- Recall: The ratio of correctly classified positive values to the total positives in a class.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8.8)$$

- F1-score: Harmonic mean (weighted average) of values given returned by recall and precision.

$$\text{F1-score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}} \quad (8.9)$$

The values of performance evaluation parameters for all the models are represented in Table 8.3.

8.5 RESULTS

In the current study, four different models – custom CNN, VGG16, ResNet-50, and InceptionV3 – were employed for predicting and classifying apple leaf diseases. The performance metrics, including classification accuracy, precision, recall, and F1-score, were used to evaluate the models [26–30]. Among these models, VGG16 emerged as the top performer, exhibiting an impressive accuracy of 85%. The precision, recall,

and F1-score for VGG16 were also commendable, with values of 0.82, 0.81, and 0.81, respectively. The values of the performance parameters of all the models are shown in Table 8.3. This suggests that VGG16 excelled in correctly identifying and categorizing apple leaf diseases, showcasing its potential for practical implementation in disease detection systems for apple orchards.

Furthermore, the comparative analysis of the models highlight the significance of choosing an appropriate deep-learning architecture. While custom CNN, ResNet-50, and InceptionV3 demonstrated respectable performance, VGG16 consistently outperformed them across all evaluation metrics. The robustness of VGG16 in handling the complexities of apple leaf images and accurately predicting diseases underscores its suitability for real-world applications. These results not only validate the effectiveness of deep learning in disease prediction but also emphasize the importance of model selection, with VGG16 standing out as the most reliable and accurate choice for the detection of apple leaf diseases in the context of Kashmir's orchards.

8.6 CONCLUSION

At the beginning of this chapter, the advancements of machine learning and deep learning and the purpose of this study were discussed. The next segment was about the work done in this regard. Following that, the experiment setup was discussed, which comprised the dataset, steps followed, and models [31–35]. The last section is about the results of the models performed using various performance evaluation parameters.

This study presents a significant advancement in the field of agriculture, specifically addressing the challenges faced by apple cultivators in Kashmir. The proposed deep learning-based system, utilizing custom CNN, VGG16, ResNet-50, and InceptionV3 models, exhibits a commendable accuracy of 85%, with VGG16 leading in performance. The classification of apple leaves into distinct categories – healthy, apple rot, *Alternaria* leaf blotch, and apple scab – proves to be an invaluable contribution to the farming community. The study not only offers an efficient and cost-effective alternative to traditional disease prediction methods but also signifies a paradigm shift toward leveraging advanced technologies for agricultural sustainability.

Looking forward, the future of this study lies in continuous refinement and expansion. Optimization of existing models, exploration of ensemble techniques, and incorporation of real-time data sources could further enhance the accuracy and applicability of the predictive system. Extending the scope beyond apple leaves to encompass a broader range of crops would amplify the impact of these models in diverse agricultural contexts. Collaborative efforts involving experts in agriculture, data science, and technology innovation could drive interdisciplinary research, fostering innovative solutions for crop disease management. The integration of Internet of Things (IoT) devices, sensors, and big data analytics holds promise for proactive disease control strategies, ensuring the resilience and prosperity of farming communities. This study marks a pivotal step toward the integration of cutting-edge technologies in agriculture, with far-reaching implications for the future of sustainable and technology-driven farming practices.

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